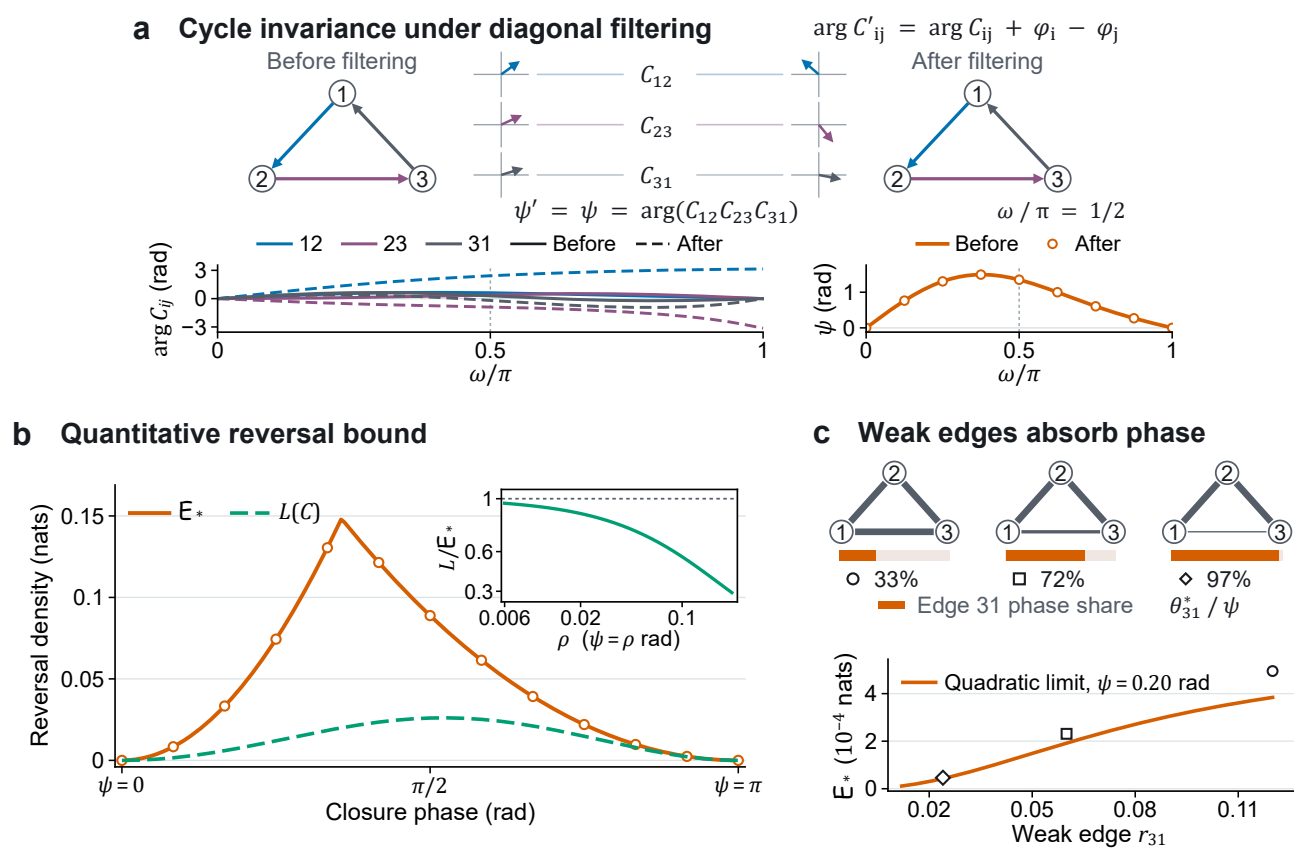

a
Cycle invariance under diagonal filtering
$\arg C'_{ij} = \arg C_{ij} + \varphi_i - \varphi_j$
Before filtering
After filtering
C_{12}
C_{23}
C_{31}
$\psi' = \psi = \arg(C_{12}C_{23}C_{31})$
$\omega / \pi = 1/2$
12
23
31
Before
After
$\arg C_{ij}$ (rad)
ψ (rad)
ω/π
b
Quantitative reversal bound
E*
$L(C)$
Reversal density (nats)
L/E^*
ρ ($\psi = \rho$ rad)
$\psi = 0$
$\pi/2$
$\psi = \pi$
Closure phase (rad)
c
Weak edges absorb phase
33%
72%
97%
Edge 31 phase share
θ^*_{31} / ψ
Quadratic limit, $\psi = 0.20$ rad
E^* (10^{-4} nats)
Weak edge r_{31}